Goethe University

Bayesian Modeling for Marketing Prof. Thomas Otter

Homework #4
Data augmentation – MNProbit

1. Draw the directed acyclic graph (DAG) corresponding to the following data-generating mechanism:
$$\epsilon \sim N\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
$$z_{1,i} = x'_{1,i}\beta + \epsilon_{1,i}, \qquad z_{2,i} = x'_{2,i}\beta + \epsilon_{2,i}, \quad z_{3,i} = x'_{3,i}\beta + \epsilon_{3,i}$$
$$y_i = \sum_{j=1}^{3} j\mathbf{I}\left(z_{j,i} = max\left(z_{1,i}, z_{2,i}, z_{3,i}\right)\right)$$
2. Write code to simulate data from this model.
3. Derive the full conditional distributions required for augmenting the z's given the data and parameter β.
4. Use the code in rbprobitGibbs.r as a starting point for a Gibbs-sampler (using data augmentation) for this model.
5. Show that your code can recover data generating parameter values when applied to simulated data.
6. Derive the full conditional distributions required for augmenting the z's given the data and parameter β for the case of correlated error terms ϵ. (Do not code this up--only derive the result analytically recalling standard multivariate normal (regression-) theory, i.e.

$$\theta = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right)$$

define

$$\Sigma^{-1} = V = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}$$

then

$$\theta_1 \mid \theta_2 \sim N\left(\mu_1 - V_{11}^{-1}V_{12}(\theta_2 - \mu_2), V_{11}^{-1}\right)$$